



Learning by Teaching: Enhancing Music Learning Through LLM-Based Teachable Agents

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Abstract. This study presents Chat Melody, an LLM-based multimodal teachable agent designed to enhance music theory learning through the Learning-by-Teaching (LBT) approach. By simulating a novice learner, Chat Melody engages students in structured dialogue that prompts explanation, error identification, and conceptual refinement. The system integrates a multimodal interface combining visual notation, auditory feedback, and AI-generated explanations. LBT is operationalized through a structured prompting strategy based on the help-seeking model, guiding progressive reasoning instead of offering direct answers. In a quasi-experimental study with 28 students, *Chat Melody* was compared to traditional teacher-guided instruction. Results showed that students using the teachable agent demonstrated greater improvements in knowledge construction. Additionally, responses to the Situational Motivation Scale (SIMS) indicated higher intrinsic motivation and perceived value in music learning. These findings underscore the potential of LLM-based teachable agents to support meaningful learning and cognitive engagement in AI-assisted music education.

Keywords: Large Language Models (LLMs) · Teachable Agent · Learning by Teaching (LBT) · Knowledge Construction · Music Education

1 Introduction

Large Language Models (LLMs) are increasingly integrated into educational environments, offering dynamic learning experiences through contextualized dialogue, adaptive feedback, and personalized instruction [12, 19]. These systems have demonstrated potential in supporting self-directed learning [11], problem-solving [16], and knowledge construction [14]. However, empirical studies suggest that students often exhibit

passive reliance on AI-generated feedback, leading to shallow engagement and limited metacognitive development [6, 18, 19]. When AI functions primarily as an information provider, learners may uncritically accept responses, hindering higher-order thinking. Addressing this challenge requires the design of AI-enhanced learning environments that promote deeper cognitive engagement and meaningful knowledge construction. A promising pedagogical approach to foster such engagement is Learning-by-Teaching (LBT), wherein students deepen understanding by teaching others. Research in cognitive science and educational psychology has shown that teaching enhances conceptual clarity, encourages reflective thinking, and promotes metacognitive regulation [5, 11, 16]. Within this framework, Teachable Agents (TAs)—AI-driven virtual learners that simulate novice students—have proven effective in domains such as mathematics, language learning, and programming [2, 4, 11, 16, 20]. Unlike traditional rule-based tutors, TAs prompt students to articulate, justify, and revise their understanding, supporting active learning and metacognitive development [17]. Recent advances in LLM-powered conversational AI have significantly expanded the potential of teachable agents. Traditional TAs, constrained by predefined rules and rigid interaction structures, often struggle to support open-ended reasoning and critical inquiry, particularly in fields like music education that demand interpretative flexibility [15]. In contrast, LLM-driven TAs leverage generative models to facilitate adaptive, context-aware, and fluid dialogues [1, 9, 11, 16]. By engaging learners in iterative explanation and reflection, LLM-powered TAs enhance the effectiveness of LBT environments, offering a more personalized and cognitively demanding learning experience [4]. Despite their growing use in programming and mathematics education, the application of LLM-powered teachable agents in music education remains underexplored. Prior studies have shown that such agents can enhance learning outcomes and self-regulation. For example, [4] implemented a ChatGPT-based TA using Gall’s help-seeking model [7], ensuring structured cognitive progression. Similarly, [11] demonstrated that concealing the agent’s knowledge state prompted learners to actively elicit and refine concepts through dialogue. In [16], an LLM-based TA was successfully used to support middle school students in mathematical concept learning by encouraging explanation and self-monitoring.

Building on these findings, this study introduces *Chat Melody*, an LLM-based tutoring system that integrates the LBT approach to support music theory learning. By combining LLM-driven dialogue with structured pedagogical scaffolding, this study explores how teachable agents can enhance learners’ motivation and promote durable knowledge construction in music education. The research questions of this study are as follows: **RQ1:** What impact does the teachable agent (*Chat Melody*) have on students’ knowledge construction in music learning? **RQ2:** How does the teachable agent influence students’ motivation in music learning?

2 The Design of an LLM-Based Teachable Agents

Our research team developed an LLM-powered teachable agent [13] to facilitate structured interactions in music learning through adaptive AI prompts (See Fig. 1). Grounded in the Learning-by-Teaching (LBT) approach, the system repositions AI from a static knowledge provider into a teachable agent that prompts learners to articulate, refine, and critically assess their understanding of music theory.

To ensure structured learning interactions, we employed a prompt-based design aligned with the five-stage help-seeking process model [7]: (1) awareness of the need for help, (2) decision to seek help, (3) identification of a potential help source, (4) strategies to elicit help, and (5) reactions to help-seeking attempts. Rather than providing direct answers, the agent simulates a novice music student to engage learners in structured reasoning, prompting them to explain concepts, justify interpretations, and refine understanding. For example, when analyzing the harmonic structure in Measure 5 of Mozart’s Piano Sonata No. 5 in G major, K. 283 (Movement 2), the agent may ask: “I see E, G, C, and G in the right hand—what does this chord represent?” This initiates the first stage of help-seeking, prompting the learner to identify the notes as a C major chord (IV in G major). The agent then asks about the G in the bass, guiding the learner to recognize the chord as a cadential six-four (IV6/4) resolving to V (D major), and ultimately to I (G major). The conversation continues with prompts such as, “What happens in the next beat?”, reinforcing the functional role of harmonic progression (IV6/4 → V → I) and deepening conceptual understanding. Through explanation and guided validation, the system transforms the LLM into a cognitive partner, supporting active knowledge construction.

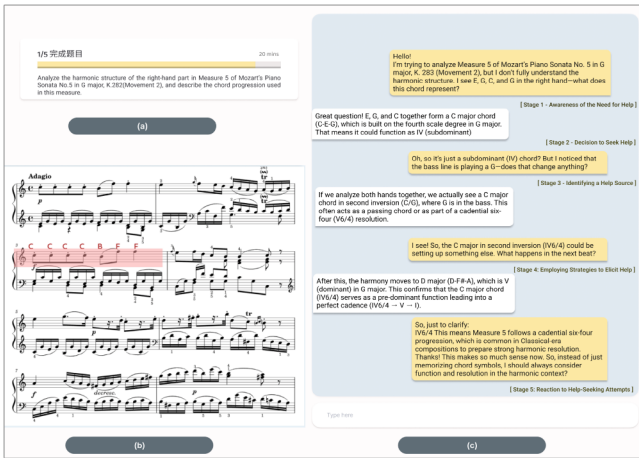


Fig. 1. Interface design of the LLM-powered Teachable Agent. The system comprises three key interactive components: (a) the problem statement area, which presents structured music analysis tasks; (b) the interactive music sheet display, which supports note-clicking and auditory feedback to connect notation with perception; (c) and the teachable agent, powered by GPT-4, which engages learners in structured dialogue as a novice learner.

Although LLMs such as GPT-4 occasionally produce inaccurate responses [10], we do not treat these errors as system flaws. Instead, following [4], we conceptualize them as learning opportunities, viewing these inaccuracies as cognitive triggers that mirror the reasoning struggles of novice learners. This perspective encourages reflection and supports knowledge construction. For instance, when the agent mislabels a harmonic function—such as misidentifying a cadential six-four as a tonic chord—the student must

diagnose the mistake, justify the correct interpretation, and explain their reasoning. This aligns with the principles of error-driven learning, in which understanding is reinforced through the detection and correction of conceptual errors. Unlike traditional AI tutors that prioritize correctness, this method promotes higher-order thinking by requiring students to critically evaluate AI-generated responses.

3 Method

- A) **Participants.** A total of 28 Chinese university students (18 females, 10 males) participated in the study, ranging in age from 19 to 29 years ($M = 24.29$, $SD = 2.62$). All had prior instrumental learning experience and self-reported a foundational understanding of music theory. Since the length of instrumental training does not necessarily reflect theory proficiency [3, 20], participants completed a 15-min pre-test on notation, scales, rhythm, chords, and harmonic structures. The test, developed by an experienced music educator, confirmed no significant difference between the experimental group ($n = 16$) and the control group ($n = 16$) at baseline ($t = 0.86$, $p = .396$). During the study, four control group participants withdrew before completing the post-test, resulting in a final sample of 12 in that group. The experimental group received instruction via a teachable agent, while the control group received teacher-guided online instruction.
- B) **Experiment Procedures and Learning Task.** The study was conducted entirely online and consisted of four phases: **Phase 1 (Pre-Test):** Before the intervention, all participants completed a pre-test assessing foundational music theory knowledge, including notation, scales, and harmonic structures. **Phase 2 (Instructional Phase):** All Participants then engaged in a two-day Zoom-based instructional phase. Day 1 focused on music notation and interpretation, covering note values, rhythm, articulation, scales, and key signatures, reinforced through guided exercises and listening activities. Day 2 addressed musical structures, particularly sonata form, using Mozart's Piano Sonata No. 5 in G Major, K. 283 (Movement 2) as the core material. Learners analyzed the exposition, development, and recapitulation, examining thematic development, modulation, and harmonic progression. **Phase 3 (Post-Test):** Following instruction, participants completed a 60-min analysis task requiring them to apply learned concepts to Mozart's piece. The experimental group interacted with Chat Melody, engaging in structured dialogue with the LLM-based teachable agent to explain, justify, and refine their analysis. The control group completed the same task independently, using only instructional materials and recorded lectures. All participants shared their screens via Zoom during the task, monitored by a researcher to prevent the use of external resources. Immediately after the analysis task, participants completed the Situational Motivation Scale (SIMS) to evaluate the impact of instructional methods on motivation. **Phase 4 (Delayed Test):** Two weeks later, a delayed test was administered to assess long-term knowledge retention.
- C) **Music Theory Knowledge: Pre-test, Post- test, and Delayed test.** We adopted pre-tests, post-tests, and delayed tests to assess participants' progress in music theory knowledge. The pre-test primarily consisted of multiple-choice questions, whereas the post-test and delayed test included multiple-choice, short-answer, analysis, and

comprehensive questions. The format remained consistent across all assessments, with variations limited to question content, answer choices, and numerical values. Each test had a total score of 50 points, with raw scores converted into standardized Z-scores on a 100-point scale. The test was developed by a faculty member in music education and a doctoral student specializing in music education assessment, both of whom have extensive teaching experience. In this study, knowledge construction is understood as a process in which individuals progressively deepen their understanding of music theory through active cognitive engagement, transitioning from basic information acquisition to more advanced conceptual comprehension [3, 18]. The delayed test evaluates long term knowledge retention [14, 18]. By administering these tests, researchers were able to more accurately assess how learners acquire and retain music theory knowledge.

- D) **Motivation Scale.** We used the Situational Motivation Scale (SIMS) proposed by [8], and adapted it to the context of music theory learning. This scale on the seven-point Likert scale evaluates four dimensions of motivation: (a) intrinsic motivation that reflects personal interest in learning music (e.g., “Because I think learning music theory is interesting”); (b) identified regulation that occurs when learners recognize the value of music education (e.g., “I am learning music because I believe it is important for me”); (c) external regulation that involves external pressures such as expectations or requirements (e.g., “I am learning music because I am supposed to”); and (d) amotivation that represents a lack of motivation or perceived relevance (e.g., “I don’t see the point of learning music theory”). This scale has been shown to have high reliability and validity in previous studies [8].

4 Results

Q1: The Impact of Chat Melody on Knowledge Construction. To assess participants’ progress in music theory knowledge, evaluations were conducted at three distinct intervals: a pre-test (before instructional intervention), a post test (immediately after the intervention), and a delayed test (two weeks later). Two ANCOVAs were performed to examine variations across these time points, thereby evaluating learners’ knowledge retention and construction. The first ANCOVA treated post-test scores as the dependent variable with pre-test scores as the covariate to determine differences in music theory understanding between the experimental group (EG) and control group (CG) following the instructional intervention. The second ANCOVA used delayed test scores as the dependent variable and post-test scores as the covariate to measure knowledge retention after a two-week period. Before conducting ANCOVA, Levene’s test was performed to verify the homogeneity of variances, with the results presented in Table 1. Both post-test and delayed test p -values were above 0.05, ensuring the validity of the equal variance assumption and confirming ANCOVA’s suitability for analysis. To verify group equivalence in prior music theory knowledge, an independent t-test was conducted on pre-test scores. The control group had a mean of 51.89 ($SD = 8.33$), and the experimental group had a mean of 48.58 ($SD = 11.14$). The result ($t = 0.86$, $p > .05$) indicated no significant difference between the groups, confirming comparable prior knowledge before instruction (see Table 2).

Table 1. The Levene's test results for knowledge construction.

Variable	Levene's test	
	<i>F</i>	<i>P</i>
Post-test for music theory knowledge	.59	.450
Delayed test for music theory knowledge	1.04	.317

Table 2. t-Test result of the pre-test scores.

		N	Mean	S.D	<i>t</i>	<i>p</i>
Pre-test	EG	16	48.58	11.14	.86	.369
	CG	12	51.89	8.33		

Table 3 presents the descriptive statistics and ANCOVA results for post-test and delayed test scores, highlighting the impact of *Chat Melody* on learners' knowledge retention and construction in music theory. The adjusted mean post-test scores were 59.65 (SE = 1.69) for the experimental group (EG) and 46.94 (SE = 2.00) for the control group (CG), with a significant difference ($F = 19.98, p < .001$), indicating greater improvement in the EG. A second ANCOVA on delayed test scores, using post-test scores as the covariate, showed adjusted means of 52.19 (SE = .93) for the EG and 45.83 (SE = 1.09) for the CG, also yielding a significant difference ($F = 17.65, p < .001$). These results suggest that learners supported by the teachable agent demonstrated stronger knowledge retention and construction than those who received teacher-guided instruction alone.

Table 3. Descriptive data and ANCOVA of the post-test results and delayed-test results.

Group		N	Mean	S.D	Adjusted Mean	S.E	<i>F</i>	<i>p</i>	η^2
Post-test	EG	16	57.30	6.72	59.65	1.69	19.98	<.001	.35
	CG	12	50.07	8.30	46.94	2.00			
Delayed-test	EG	16	55.25	6.88	52.19	.93	17.75	<.001	.11
	CG	12	41.75	9.55	45.83	1.09			

Q2: Influence of the Teachable Agent on Learners' Motivation. The results of the motivational scale data (see Table 4) indicate that the use of Chat Melody in music theory learning had a significant impact on learners' intrinsic motivation and identified regulation, while no significant differences were found in external regulation or amotivation between the experimental and control groups. Specifically, learners who engaged with the teachable agent reported significantly higher intrinsic motivation ($t = 2.55, p < .05$, Cohen's $d = .97$) and identified regulation ($t = 2.81, p < .01$, Cohen's $d = 1.07$) compared to those who received teacher-guided online instruction. In contrast, the findings

revealed no significant effect of Chat Melody on external regulation ($t = .61, p = .546$, Cohen's $d = .23$) or amotivation ($t = -0.77, p = .446$, Cohen's $d = -0.30$).

Table 4. Independent Sample t test for motivation.

Group		Mean	S.D.	S.E.	<i>T</i>	<i>p</i>	<i>Effect Size</i>
Intrinsic motivation	EG	25.56	3.67	.92	2.55	.017	.97
	CG	21.83	4.04	1.17			
Identified regulation	EG	25.38	3.24	.81	2.81	.009	1.07
	CG	21.25	4.54	1.31			
External regulation	EG	16.13	4.91	1.23	0.61	.546	.23
	CG	15.00	4.67	1.35			
Amotivation	EG	10.25	5.05	1.26	-0.77	.446	-0.30
	CG	11.75	5.12	1.48			

5 Discussion and Conclusion

This study investigated how an LLM-powered teachable agent, *Chat Melody*, supports music theory learning through the Learning-by-Teaching (LBT) approach. Results showed that learners using Chat Melody achieved significantly greater post-test and delayed-test gains than those receiving teacher-guided instruction, indicating improved knowledge acquisition and sustained knowledge construction ($\eta^2 = .35$; $\eta^2 = .11$). The agent also enhanced intrinsic motivation and identified regulation, suggesting that simulating a novice learner encouraged students to take ownership of learning and engage more deeply. No significant effects were found for external regulation or amotivation, highlighting the agent's specific impact on self-directed motivation. While limited by a small sample and lack of interaction data, the study offers evidence that LLM-based teachable agents can promote meaningful, self-motivated learning in music education. Future research should incorporate qualitative methods and broader populations to explore the cognitive and emotional dynamics of human-AI collaboration in learning contexts.

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